

No External-Field Effect: Cluster-Satellite Lensing, the Cassini Quadrupole and the Outer Solar System Point to an EFE-Blind MOND Sector

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Abstract

In modified-gravity versions of MOND a system’s internal boost depends on the external gravitational field it sits in (the external-field effect, EFE). We test the quadrature law $g^2 = g_N^2 + a_0 g_{\text{ext}}$ with $a_0 = \frac{1}{2}c\sqrt{G\rho_\Lambda}$ against three data sets that are sensitive to the EFE in different ways. (i) Weak lensing of satellite galaxies in massive clusters (Sifón et al. 2018, five stellar-mass bins) is fitted by the full, isolated boost with $\chi^2 = 0.7$ for 5 bins and a mean offset of +0.01 dex; the same law with the cluster’s field included, $\eta = g_{\text{ext}}/a_0 = 0.59\text{--}0.81$, fails by $\chi^2 = 99.7$, and a direction-blind rule that keeps only the field’s magnitude fails equally ($\chi^2 = 99.5$). (ii) The anisotropic part of the EFE predicts a Cassini quadrupole $Q_2 = 2.195 \times 10^{-26} \text{ s}^{-2}$, 6.3σ above the measured value. (iii) In the outer Solar System the Galactic EFE pumps the perihelia of detached trans-Neptunian objects, while a central force conserves them; the pre-registered verdict of that secular run was inconclusive. The data therefore ask for a MOND sector whose boost is computed from a bound system’s own field alone. This costs two results that preferred an EFE, the Milky Way’s Gaia DR3 outer decline and Chae’s rotation-curve signal, and it makes a sharp prediction for Gaia DR4: wide binaries boosted as isolated MOND systems, $\gamma_v = \sqrt{\nu} \approx 1.20$ at 10 kAU and 1.51 at 20 kAU for $1.5 M_\odot$. No covariant theory with this property exists yet.

1 Cluster-satellite lensing

Sifón et al. (2018) measured subhalo masses m_{bg} of satellites in MENeCS clusters ($z < 0.15$; median dynamical host mass $M_{200} \sim 6 \times 10^{14} M_\odot$) inside r_{bg} , where the subhalo density equals the host’s local density. We read their table directly (five bins, $\log M_* = 9.51\text{--}11.01$) and estimate $r_{\text{bg}} = 36\text{--}149 \text{ kpc}$ by matching an isothermal subhalo to an NFW host ($M_{200} = 6 \times 10^{14} M_\odot$, $c = 4$). Each satellite’s baryons are $1.2 M_*$; we add the galaxy-level retained cold mass found in early-type galaxies (13% of the cosmic share), which contributes about 5%. At r_{bg} the internal field is tiny, $x = g_N/a_0 = 0.004\text{--}0.008$, while the cluster’s field is $\eta = 0.59\text{--}0.81$.

reading	χ^2 (5 bins)	mean offset (dex)
isolated boost $\nu(x)$ (no EFE)	0.7	+0.01
magnitude EFE $\nu(\sqrt{x^2 + \eta^2})$	99.5	–
linear EFE $\nu(x + \eta)$	99.7	+0.81
no phantom in a bound satellite (“ownership”)	131.6	+0.93

Any rule that lets the cluster’s field enter the satellite’s boost caps ν near 1.5 instead of 11–15. Λ CDM also fits these data (it is the paper’s model), so this test separates readings of MOND, not MOND from Λ CDM.

2 The Cassini quadrupole and the outer Solar System

Solving the QUMOND field equation for the Sun in the Galactic field (multipole solver, observed $g_{\text{ext}} = 2.146 \times 10^{-10} \text{ m s}^{-2}$) gives $Q_2 = 2.195 \times 10^{-26} \text{ s}^{-2}$, 6.3σ above Cassini’s $(3 \pm 3) \times 10^{-27}$ (Hees et al. 2014); Cassini alone would require $\kappa < 0.171$ (2σ). With the EFE removed $Q_2 = 0$. In a secular model of detached trans-Neptunian objects (giant-planet quadrupole, Galactic tide, the QUMOND anomaly), the EFE’s quadrupole drives large perihelion changes. With the EFE, 57% of objects are driven below $q = 25 \text{ AU}$ and the detached occupancy is 0.195, against 19% and 0.067 for Newton. A control run with only the central monopole (planets and tide off) conserves perihelia to 0.005 AU over 200 Myr, as a central force must. That run’s pre-registered verdict was inconclusive: the numbers are small and Neptune scattering is not modelled. The EFE is what makes MOND overpopulate the detached disc (Vokrouhlický, Nesvorný & Tremaine 2024).

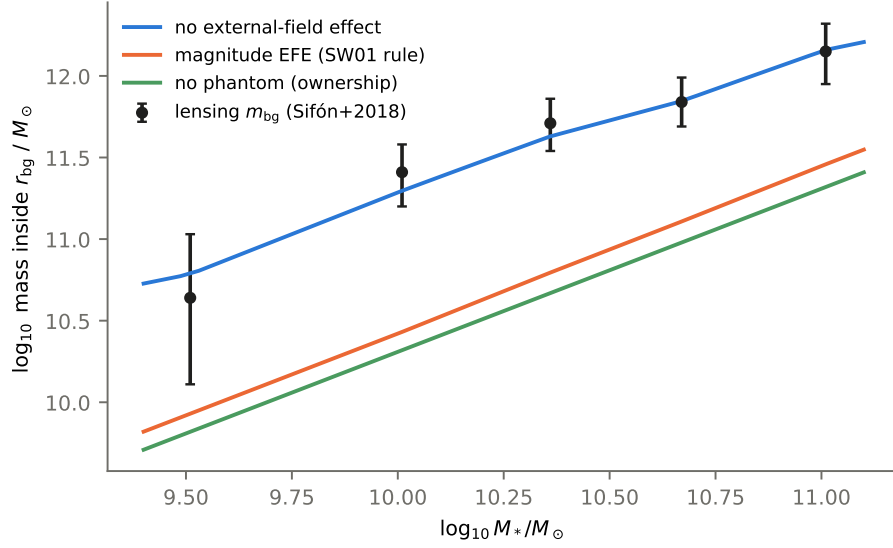


Figure 1: Satellite masses inside r_{bg} (points) against three readings of the law.

3 What an EFE-free rule costs and predicts

Two results in this programme preferred an EFE: the Milky Way’s Gaia DR3 outer decline and Chae’s rotation-curve EFE signal. For the first, the isolated kernel (ν_{RAR} in that lane) is $2.3\text{--}2.4\sigma$ too shallow nominally and up to 2.5σ for the least favourable (shallowest) baryonic model. A 1-D LMC field model reduces this to $1.1\text{--}1.6\sigma$, and disequilibrium is not modelled. For the second, refitted with this law, it is 1.7σ on the canonical footing and 2.7σ on the alternative (2.2 and 3.0σ with ν_{mono}). WALLABY cannot decide: its directional-EFE stack gives an amplitude of -1.70 on a scale where AQUAL is $+1$, with error ± 2.98 (analytic; ± 2.12 bootstrap), i.e. 0.6σ from zero and 0.9σ from AQUAL; the run is consistent with AQUAL and with no EFE alike. An EFE-free rule gives up both. In exchange it predicts wide binaries boosted as isolated systems: $\gamma_v = \sqrt{\nu(y)} = 1.06, 1.20, 1.51$ at 5, 10, 20 kAU for $1.5 M_\odot$ (3-D separation; the registered statistic is not recomputed here), against 1.11 at 20 kAU for the magnitude rule and 1 for Newton (Fig. 2). Gaia DR4 (December 2026) decides; a Newtonian result would exclude this reading.

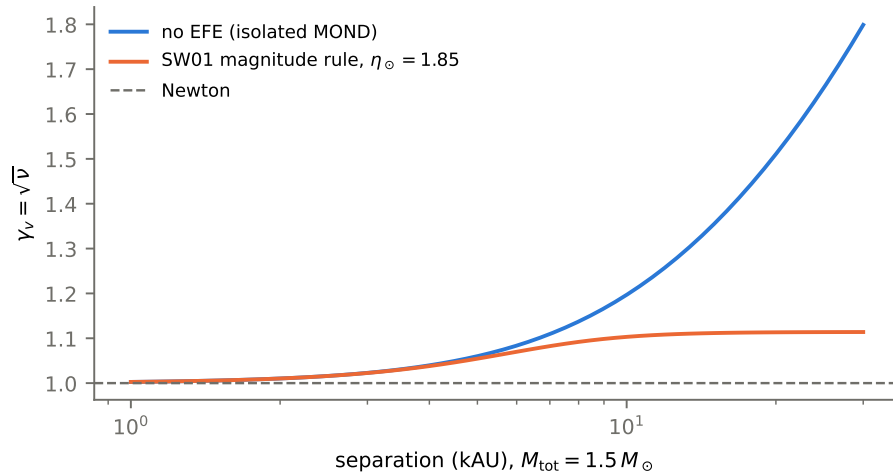


Figure 2: Wide-binary velocity boost for the EFE-free rule, the magnitude-only rule and Newton.

4 Summary

Satellite lensing, the Cassini quadrupole and the outer Solar System each disfavour an external-field effect, the first even in its direction-blind, magnitude-only form. The consistent reading is a MOND sector computed from each bound system’s own field. No covariant action with this property is known; the programme’s direction-blind construction keeps the field’s magnitude and is excluded by the satellites. The test is Gaia DR4. The coefficient $\kappa = \frac{1}{2}$ remains fitted; the cold mass of cosmology is still required.

Reproducibility. `sonnet55_push/cold_mass/cm14b_sifon_table.py`, `cm15_sw01_rule_vs_satellites.py`, `cm08_etg_retained_fraction.py`; `sonnet55_push/puzzle_32pi/p58_cassini_kappa_bound.py`, `p57_etno_secular_nbody.py`, `qumond_efe_multipole.py`; `fetch log data_assembly/sifon2018/`; `figures make_paper44_figures.py`; `audit PAPER44_audit.py`.

References. Sifón C. et al., MNRAS 478, 1244 (2018), arXiv:1706.06125. Hees A., Folkner W. M., Jacobson R. A., Park R. S., Phys. Rev. D 89, 102002 (2014). Vokrouhlický D., Nesvorný D., Tremaine S., ApJ 968 (2024), arXiv:2403.09555. Milgrom M., Phys. Lett. A 253, 273 (1999).